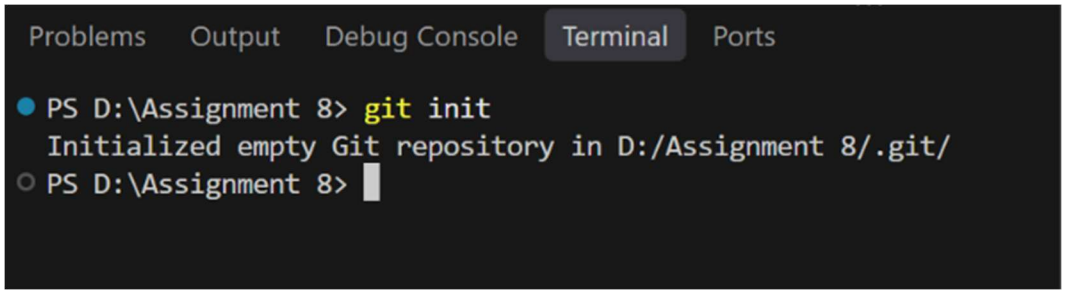
Assignment 8 - Git Basics

TW Lab 2025

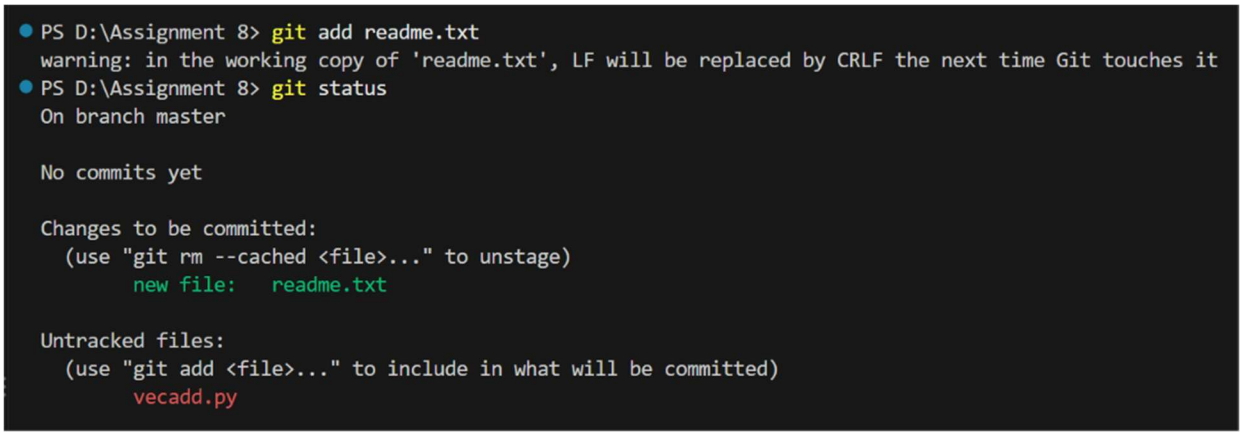
Name: Saarvik Singh Suryavanshi
ID: 2025UCP1681
Batch: A3

Step 1: Initialising the repository (git init)



I moved into the Assignment 8 folder and ran **git init**. Git created an empty repository in the folder, which is confirmed by the message shown. I skipped step 2 since I already have the setup of my Github username and email.

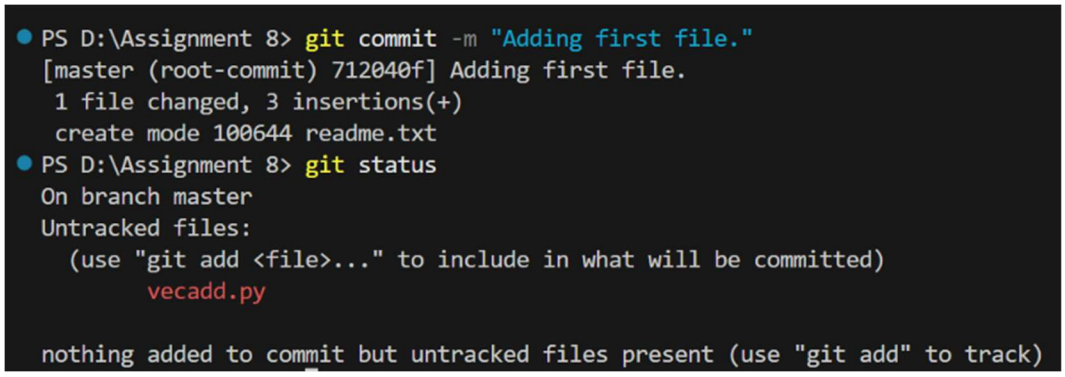
Step 3: Stage readme.txt and check status (git add, git status)



I created readme.txt and staged it with **git add readme.txt**.

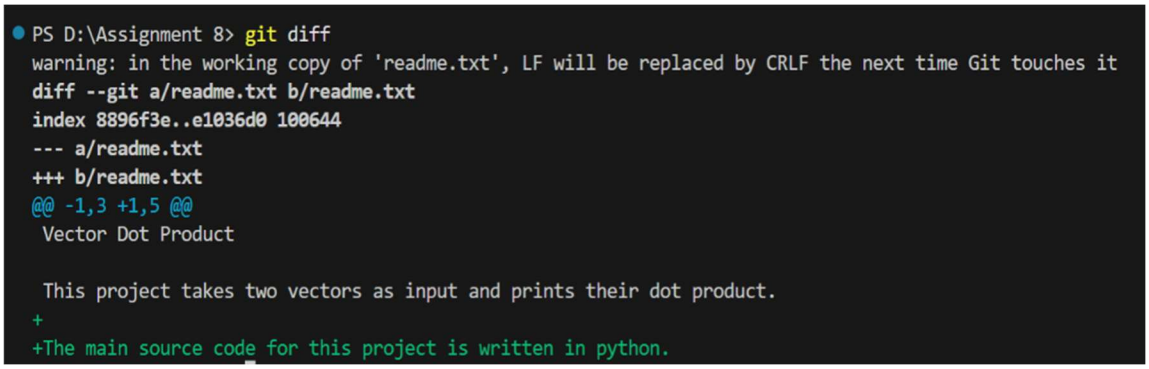
git status then listed readme.txt as a new file to be committed, while vecadd.py was still untracked.

Step 4: Commit the file (git commit)



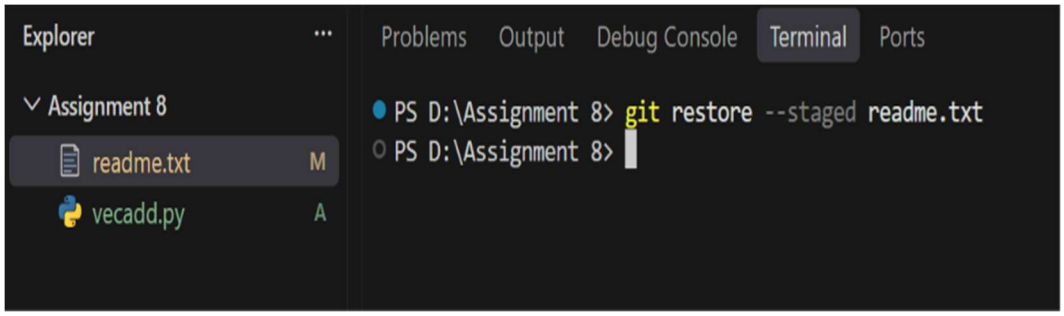
I committed the staged file with the message "Adding first file." This created the first (root) commit, 712040f. Running **git status** again showed that only vecadd.py remains untracked, so readme.txt is fully committed.

Step 5: View changes (git diff)



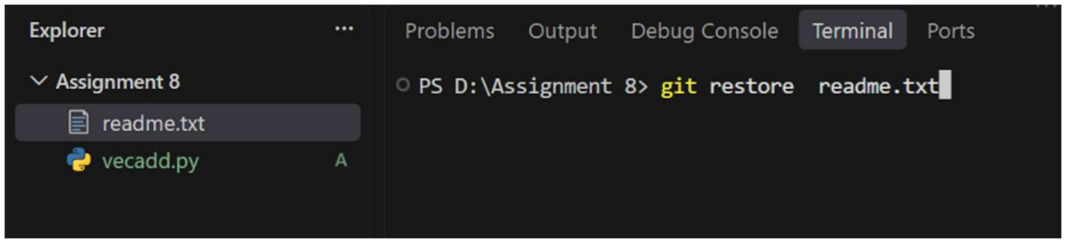
I added one new line to readme.txt. **git diff** compared the working copy with the last commit, showing the added lines in green with a + sign.

Step 6a: Restore at the staged stage



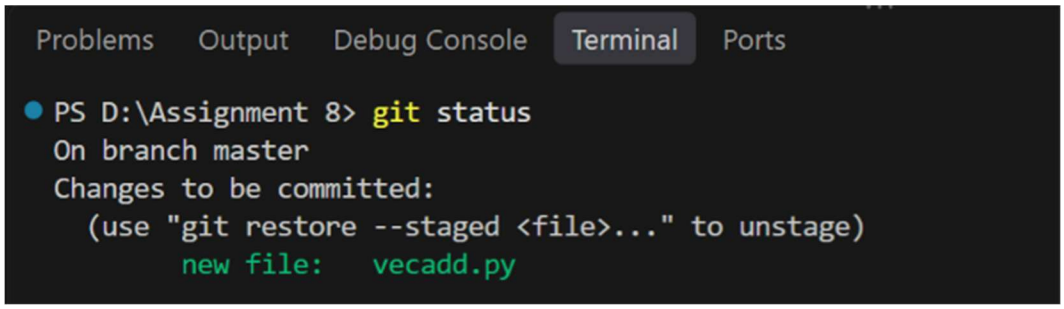
After staging the modified readme.txt, I ran **git restore --staged readme.txt**. This only unstaged the file; my edits stayed in the file, as the M (modified) mark in the Explorer shows.

Step 6b: Restore at the modified stage



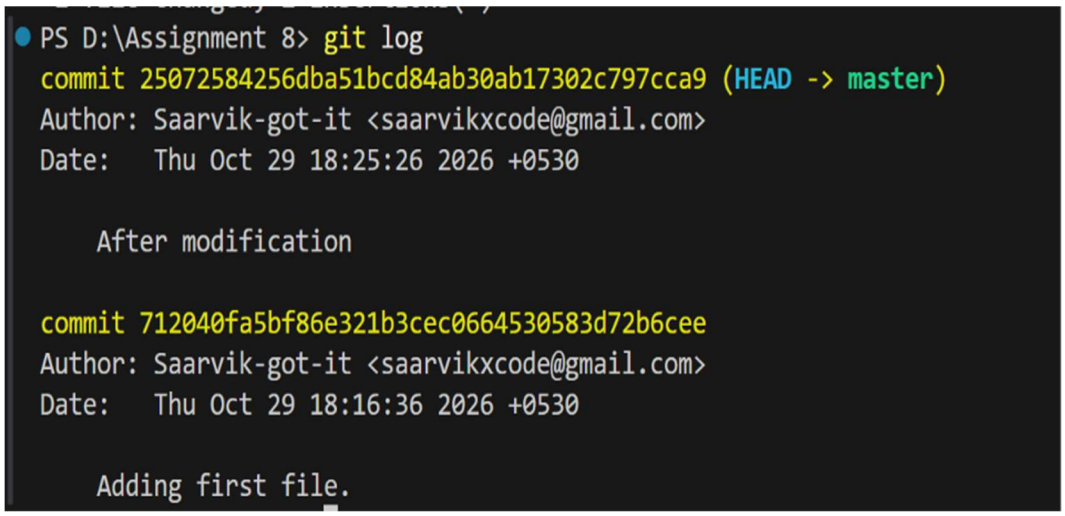
I then ran **git restore readme.txt** on the unstaged file. This brought the file back to its last committed version, so the M mark disappeared from the Explorer.

Step 6c: Status after restoring



git status now shows only vecadd.py as a new file waiting to be committed. The readme.txt changes were removed by the restore.

Step 6d: Commit again and view history (git commit, git log)



I modified readme.txt again and committed it with the message "After modification". **git log** now shows two commits, with the newest at the top and HEAD pointing to master.

Step 8: Revert the previous commit (git revert)

```
PS D:\Assignment 8> git revert  25072584256dba51bcd84ab30ab17302c797cca9
[master 8ec6f99] Revert "After modification"
 1 file changed, 2 deletions(-)
PS D:\Assignment 8> git log
commit 8ec6f9923a3cbcd0e60d1fc0e65a3cd455ee1567 (HEAD -> master)
Author: Saarvik-got-it <saarvikxcode@gmail.com>
Date:   Thu Oct 29 18:26:04 2026 +0530

    Revert "After modification"

    This reverts commit 25072584256dba51bcd84ab30ab17302c797cca9.

commit 25072584256dba51bcd84ab30ab17302c797cca9
Author: Saarvik-got-it <saarvikxcode@gmail.com>
Date:   Thu Oct 29 18:25:26 2026 +0530

    After modification

commit 712040fa5bf86e321b3cec0664530583d72b6cee
Author: Saarvik-got-it <saarvikxcode@gmail.com>
Date:   Thu Oct 29 18:16:36 2026 +0530

    Adding first file.
```

I reverted the "After modification" commit using its hash. Git did not delete it. Instead it created a new commit, "Revert After modification", so all three commits are visible in the log.

Step 9: Commit vecadd.py (git add, git commit, git log)

```
PS D:\Assignment 8> git add .
warning: in the working copy of 'vecadd.py', LF will be replaced by CRLF the next time Git touches it
PS D:\Assignment 8> git commit -m "Added vecadd.py"
[master 2b0354f] Added vecadd.py
 1 file changed, 8 insertions(+)
 create mode 100644 vecadd.py
PS D:\Assignment 8> git log
commit 2b0354fb952293f5731d2ac3d1db1e6a1c166062 (HEAD -> master)
Author: Saarvik-got-it <saarvikxcode@gmail.com>
Date:   Thu Oct 29 18:28:44 2026 +0530

    Added vecadd.py

commit 8ec6f9923a3cbcd0e60d1fc0e65a3cd455ee1567
Author: Saarvik-got-it <saarvikxcode@gmail.com>
Date:   Thu Oct 29 18:26:04 2026 +0530

    Revert "After modification"

    This reverts commit 25072584256dba51bcd84ab30ab17302c797cca9.

commit 25072584256dba51bcd84ab30ab17302c797cca9
Author: Saarvik-got-it <saarvikxcode@gmail.com>
Date:   Thu Oct 29 18:25:26 2026 +0530

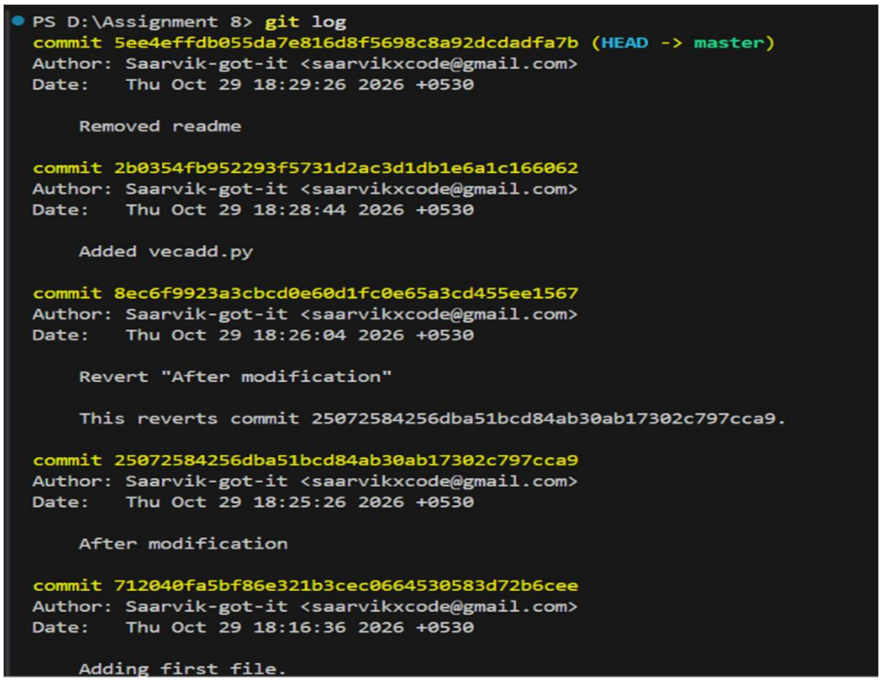
    After modification

commit 712040fa5bf86e321b3cec0664530583d72b6cee
Author: Saarvik-got-it <saarvikxcode@gmail.com>
Date:   Thu Oct 29 18:16:36 2026 +0530

    Adding first file.
```

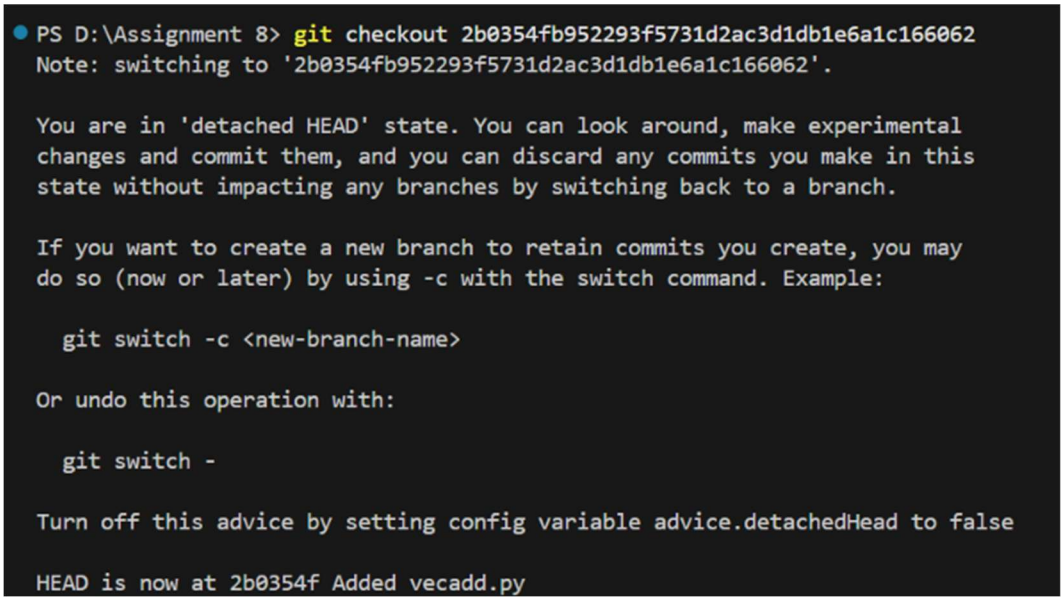
I staged vecadd.py with **git add .** and committed it as "Added vecadd.py". The git log output shows this new commit at the top of the history.

Step 10: Remove readme.txt (git rm, git commit, git log)



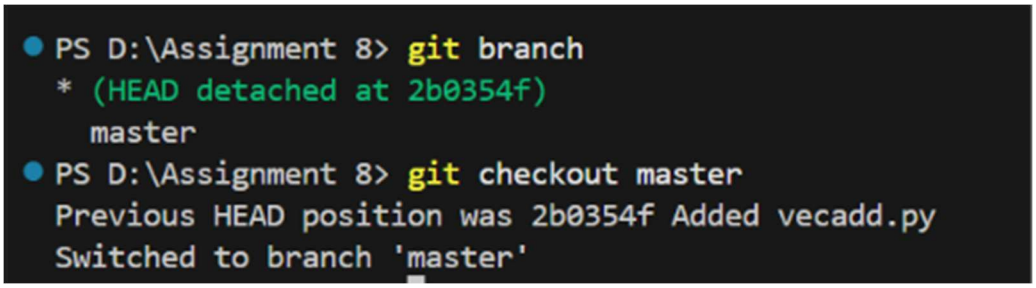
I removed readme.txt with **git rm** and committed the removal as "Removed readme". The log now has this commit at the top, and the old commits are still kept in the history.

Step 11: Checkout an earlier commit (git checkout <commit-hash>)



I checked out the earlier commit 2b0354f ("Added vecadd.py"). Git moved , and the working directory now shows the project as it was at that commit.

Step 12: Return to the latest commit (git checkout master)



git branch confirmed that HEAD was detached at 2b0354f. I then ran **git checkout master** and Git switched back to the master branch, returning to the latest commit.
